A System with an ...

Description

The objective of this project is to take advantage of emerging and existing ……. technologies to develop a system that is targeted towards …

To achieve these aims, this project will be implemented on or with …

The system will provide . . .

. . . as well as providing interactivity and social aspects to enhance their experience.

The project will choose an appropriate architecture . . .

The objective of this project is to take advantage of existing and emerging AI, AR technologies, and smart sensors to develop a system that is targeted towards people who want safer, more personalised skincare, makeup, hair care, nail care, fragrances, and men's grooming routines.

To achieve these aims, this project will be implemented as a mobile app, combining features like AI-driven skin analysis, ingredient safety checks, AR makeup try-ons, personalised recommendations, multi-criteria product comparison, and smart sensor integration.

The system will provide ways for consumers to:

- Check if ingredients in products are safe by typing them in, scanning barcodes, or taking a picture of the product.
- Get personalised routines for skincare, haircare, nail care, makeup, fragrances, or men's grooming products based on their skin, hair, or nail type, conditions (like acne or dryness), or preferences, using built-in AI analysis of uploaded selfies or sensor data (e.g. smart watches).

- Use a dermatologist mode to share scans and routines with real dermatologists for professional guidance.
- Detect allergies by automatically flagging products containing ingredients consumers are allergic to.
- Try on makeup virtually or through smart mirrors before buying, including foundation, lipstick, eyeshadow, and more.
- Compare two or more products using a multi-criteria system considering price, brand reputation, eco-friendliness, consumer ratings, ingredient safety, and effectiveness for their specific condition, and see a match percentage for each product.
- Apart from filtering by product type, filter products based on sustainability, cruelty-free, vegan, or recyclable packaging.
- Receive suggestions for safer or better alternatives if a product doesn't match their needs.

The app will also include interactive and motivational features like:

- Quizzes such as "Find your skin type in a few clicks" or "Build your routine in 3 steps."
- AI-generated before-and-after progress pictures to show potential improvements over weeks or months.
- Gamification elements like daily challenges, streaks, or badges for following routines consistently.
- A personalised skin coach that checks in daily with advice like, "Your skin looks a little dry today, maybe skip the exfoliator and use more moisturiser."

Social and community features will enhance engagement, such as:

- Sharing skincare or grooming routines, "look of the day," or product reviews.

- Routine swap feature to see what products people with similar skin types are using.
- Option to follow influencers or dermatologists for verified tips.
- Community leaderboards for completing quizzes, sharing reviews, or participating in challenges.

The project will choose a cloud-based architecture to support AI analysis, AR try-ons, product databases, recommendations, and sensor integration, allowing the app to scale efficiently and stay reliable.

Requirements Engineering

Feasibility Study

1. Are there similar systems to this in the marketplace (in Desktop, Cloud, Android, iOS, Windows …)?

   Identify at least 4 examples of Similar Systems.

   Describe each of them, by gathering information from their websites, "*put directly copied material in quotes and italics*". The more information gathered here will reduce the time required to go back to the websites. Reference the URL for each.

   1) https://www.paulaschoice-eu.com/
   Paula's Choice has a bunch of ways to shop for skincare, whether it's by skin concern (like acne, dull skin), by routine step (cleansers, exfoliants, moisturizers), or by ingredient (like retinol or vitamin C). For people who are new to skincare, there's also a big educational side to the website. In the Beautypedia section you can find articles, guides, and even e-courses. They also have a skincare ingredient checker and dictionary where you can learn more about what different ingredients actually do and how good they are. If you're unsure about your own skin type, they explain how to identify it and then suggest products that are targeted for your specific needs. On top of that, you can also take their quick quiz to "Find your routine in just 3 clicks."

   2) https://skinsort.com/
   SkinSort lets you "See what's inside any product" just by searching the product name or even uploading a picture of it. The website has two main sections: Explore and Tools. In Explore, everything is split into categories like Skincare, Makeup, Haircare, and Bath & Body, and each of those breaks down into specific product types. The Tools section is probably the most useful part, as it includes things like Ingredient Checker, Routine Checker, Compare Products, Skincare Dupes, Fungal Acne Checker, and even an Ingredient Dictionary. Overall, it feels like a really complete experience if you want to analyse products, build a routine, or just find alternatives that might suit your skin better.

   3) https://www.laroche-posay.co.uk/en_GB/spotscan.htm
   La Roche-Posay is a really popular skincare brand, and their website has a few different services to help you figure out the right routine. They offer a quiz to match products to your skin type, but what really stands out is their AI tool called "Spotscan+." It's described as "a free skin analysis tool powered by artificial intelligence and co-developed with dermatologists." What makes it cool is that it actually gives an "instant analysis for blemish-prone skin" by comparing your selfies to "6,000 scientific images of all phototypes with different levels of blemishes." All you have to do is take three selfies, and then the tool analyzes your skin and gives you a personalised skincare routine. I think it's more advanced than most other

quizzes because it feels closer to an actual dermatologist consultation, just online and completely free.

4) https://www.kikocosmetics.com/en-ie/beauty-services/foundation-finder/
KIKO Milano is a pretty well-known makeup brand, but what I thought was interesting is the way they use AI tools on their website. They have an "Instant Makeup Try-On" feature and also a "Find My Perfect Shade" tool. The shade finder starts out like a quiz, where you answer questions about your skin type and features, and then it asks you to upload a photo of your face. From there, the system analyses your skin and helps match you with the right foundation shade. After that, you can actually try it on virtually using your camera, which makes it way more interactive than just reading product descriptions. What's cool is that the virtual try-on isn't just for foundation. It lets you test out eyeshadow, lipsticks, and glosses with just a few clicks, so you can see how something bold, like a red lipstick or a bright blue eyeshadow, would actually look on you before buying it.

2. Identify the main system features and services provided in the reviewed systems, above. Consider the existing systems and the services they provide. [Reference and copy the URLs]

1) https://www.paulaschoice-eu.com/
- Shopping options organised by skin concern, routine step, or ingredient.
- Educational section (Beautypedia) with articles, e-courses, ingredient checkers, and ingredient dictionaries.
- Multiple quizzes like "Find your routine in just 3 clicks" or "Find out your skin type" to help consumers identify their skin type and choose the right products.

2) https://skinsort.com/
- Consumers can search or upload the image of a product to see detailed information (ingredients, explanations, usage, reviews, and alternatives).
- Ability to compare two products side by side.
- Useful tools: Ingredient Checker, Routine Creator, Compare Products, Skincare Dupes, Fungal Acne Checker, Ingredients Dictionary.
- Multiple categories:
    - Skincare: cleansers, masks, treatments, moisturisers, eye care, sun care, lip care.
    - Haircare: shampoos, conditioners, treatments, styling.
    - Makeup: face, eye, lip, and tools.
    - Bath & Body: hand & foot, bath & shower, body moisturisers, body care, other beauty.
- "Explore" section for browsing and discovering products across all categories.

3) https://www.laroche-posay.co.uk/en_GB/spotscan.html

- AI-powered skin analysis tool ("Spotscan+") that recommends a personalised skincare routine based on the results.
- "Find my routine" quiz to recommend personalised products.
- Coaching App which provides daily check-ins to establish consistent skincare habits.
- Online Skin Consulation service for tailored expert advice.

4) https://www.kikocosmetics.com/en-ie/beauty-services/foundation-finder/

- AI-powered tools for makeup:
  - Foundation Finder: quiz questions and photo upload to analyse skin and suggest the right foundation shade.
  - Instant Makeup Try-On: virtual AR try-on using the camera to see how products look in real time.
  - Skincare Pro Advisor: quiz questions and photo upload to analyse skin and suggest the right skincare products.

3. In what ways would consumers have accomplished the activity (get/use information), when not using a system, an app or online services?

Before the existence of apps and online services, consumers had to rely on more traditional, offline methods to get information about skincare, makeup, haircare, or other beauty products. Some of the most common ways included:

- Visiting dermatologists or beauty consultants in person to get advice about their skin type, conditions, or recommended routines.
- Going to physical stores or pharmacies, where beauty advisors would give product suggestions, explain how to use them, and sometimes provide samples.
- Reading magazines, books, or brochures about skincare and beauty trends.
- Asking friends, family, or colleagues for recommendations and product experiences.
- Trying out different products to see what worked and what didn't, often without much background knowledge.
- TV or printed advertisements, which were one of the few ways to learn about new products and their supposed benefits.

These methods were often time-consuming, expensive, and not personalised. For example, in-person consultations weren't always affordable or accessible, magazines didn't cover every individual's needs, and trial-and-error could be expensive and even harmful for sensitive skin. This shows why digital systems and apps became so valuable: they make the process faster, cheaper, more interactive, and much more tailored to each person.

4. Describe a new type of system.The proposed system
Explain in detail how it might operate for different end consumers.
Consider the existing systems that provide similar services to different end consumers and system adminstrators.

Take inspiration from the systems identified in section 1 and key features identified in section 2.

The proposed system is a mobile app called PerfectMatch, designed to provide consumers with safer, more personalised beauty and self-care routines across multiple categories: skincare, makeup, hair care, nail care, fragrances, and men's grooming. It takes inspiration from existing systems such as Paula's Choice, SkinSort, La Roche-Posay, and KIKO Milano, but combines their strongest features into one single platform while also expanding with new tools for interactivity, sustainability, and professional support.

How the system operates for different end consumers:

- General consumers:
  - Can scan or upload product information (via barcode or photo) to check ingredients and learn whether the product is safe, effective, or suitable for their skin, hair, or nails.
  - Receive personalised routines generated by the system's built-in AI analysis, based on selfies, quizzes, or smart sensor data (e.g. skin moisture from smartwatches).
  - Use AR features (e.g. smart mirrors or real time camera) to try on makeup virtually, test foundation shades, or preview how skincare might improve their skin over time through AI-based progress visualisations.
  - Compare products using a multi-criteria comparison system, which shows not only ingredient safety but also brand reputation, price, eco-friendliness, and even a match percentage to their personal needs.
  - Access filters for eco-friendly, cruelty-free, or vegan products.
  - Stay motivated with gamification features like challenges, streaks, and badges for sticking to routines.
- Dermatologists (via Dermatologist Mode):
  - Consumers can opt in to share selfies, scans, and routines directly with verified dermatologists through the app.
  - Dermatologists can provide feedback and adjustments to routines, making the system more credible and medically reliable.
- Influencers and community members:
  - Influencers and beauty enthusiasts can share routines, product reviews, and "look of the day" posts, creating an interactive community space.
  - The "routine swap" feature allows consumers to see what others with similar skin or hair types are using, encouraging community learning.
  - Community leaderboards encourage engagement by rewarding activities, such as completing quizzes, posting reviews, or joining challenges.

5. Who are the stakeholders? How would this new system affect them positively or negatively?

The main stakeholders in PerfectMatch are:

1. General consumers/consumers
    - Positive effects: Gain access to personalised routines, safe product recommendations, ingredient checks, AR try-ons, and dermatologist guidance all in one place. This increases confidence when buying products, reduces trial-and-error, and can lead to better skin, hair, or nail health. It may also save money by reducing the need for paid skin consulations.
    - Negative effects: Consumers could become overly reliant on the app instead of seeking professional medical advice when needed. Privacy concerns may arise when uploading selfies or sharing personal data.
2. Dermatologists and medical professionals
    - Positive effects: A new digital channel to connect with clients, and potentially expand their reach to people who wouldn't normally book consultations. The dermatologist mode also increases credibility of the app.
    - Negative effects: Extra workload for dermatologists if too many consumers seek consultations through the platform.
3. Influencers and beauty content creators
    - Positive effects: A platform to share routines, build communities, and engage with followers. Opportunities for collaborations with brands or the app itself.
    - Negative effects: Risk of spreading misleading or biased product promotions.
4. Beauty and skincare brands
    - Positive effects: Increased visibility for their products within a wider market, especially if recommended by the app as "safe" or "best match."
    - Negative effects: Brands whose products contain questionable ingredients may be flagged negatively, which could harm their reputation and sales.
5. System administrators/developers
    - Positive effects: Gain valuable experience in managing AI-driven, AR-integrated, and community-focused systems. They can monetise through partnerships, ads, or affiliate sales.
    - Negative effects: Responsible for constant updates, moderation, and handling sensitive consumer data securely. A failure in this area could lead to trust issues or legal problems.
6. Retailers/e-commerce platforms
    - Positive effects: Integration with the app can drive product sales by making purchasing easier once consumers have made a decision.
    - Negative effects: Competition between platforms, as consumers might prefer buying directly through PerfectMatch instead of visiting brand-specific websites.

6. What other research would be necessary to ascertain feasibility, Market Research information for marktet size e.g., ownership of smartphones,

number and age of people in the market sector targeted by the system …? (Gartner Research, Cental Statisitics Office etc.)

Category	**What does it entail?**	**Reseacrh results**
Market size and demographics	Collect data on the number of potential consumers within the target sector, including age ranges, gender, and geographic regions most likely to use the app.	According to GSMA Intelligence [1], smartphone adoption across Europe is already above 85% and expected to continue growing steadily, with particularly high usage in the 18–34 age bracket. This suggests that the majority of the target audience (young adults interested in skincare and beauty) are reachable via mobile platforms.
Technology adoption	Investigate smartphone ownership rates, internet access, and operating system distribution (Android vs. iOS). Sources such as national statistics offices (e.g. Central Statistics Office) would provide reliable insights into device and app usage habits.	Recent Pew Research Center surveys [2] show that smartphone ownership and daily internet access are now widespread across both developed and emerging economies, with ownership rates exceeding 90% among 18–29 year olds. This supports the feasibility of deploying the system primarily as a mobile app on iOS and Android.
Consumer behaviour trends	Identify how often consumers purchase skincare, makeup, haircare, or grooming products, and to what extent they currently use online tools, apps, or social media for product discovery and decision-making.	Market research reports [3] indicate that skincare and beauty purchases are increasingly shifting online, with over 40% of consumers purchasing skincare digitally at least once a month. In addition, social media and apps are becoming a primary source of product discovery, which validates the inclusion of AR try-ons, quizzes, and community sharing features in the proposed system.

Competitor analysis	Examine existing apps and platforms in this sector to evaluate their popularity, pricing models, and consumer engagement levels. This would help highlight gaps in the market that the proposed system could address.	Reviewing existing tools such as Paula's Choice Beautypedia, and La Roche-Posay's Spotscan+ [4] confirms that AI-based skin analysis and virtual try-on features are already in use, but no single system combines these functions with ingredient safety checks, gamification, and professional dermatologist input in one platform. This indicates a viable gap in the market.
Privacy and data security	Research consumer attitudes towards sharing sensitive data such as selfies, skin scans, or ingredient preferences.	Recent studies [5] highlight that consumers are increasingly cautious about sharing sensitive personal data, such as selfies or health information, especially when AI analysis is involved. This reinforces the need for GDPR compliance, transparent disclaimers, and ethical data practices in the app's design.
Partnership opportunities	Explore the willingness of skincare brands, dermatologists, or beauty influencers to collaborate with such a system.	Industry reports show growing collaborations between beauty brands, dermatologists, and digital platforms. For example, La Roche-Posay's AI Spotscan+ [6] was co-developed with dermatologists. For PerfectMatch, partnerships with dermatologists and sustainable beauty brands would not only improve credibility but also help address potential gaps in product databases.

REFERENCES

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[3] “Skincare Statistics 2025 By Gender, Concerns, Conditions”, 2025, Trishita Deb, Market.us, https://media.market.us/skincare-statistics/
[4] https://www.laroche-posay.co.uk/en_GB/spotscan.html , https://www.paulaschoice-eu.com/
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[6] https://www.laroche-posay.co.uk/en_GB/spotscan.html

7. Make an initial list of functional and non-functional requirements for the new system.

Functional Requirements:

1. Consumers can create personal accounts and manage profiles, including skin, hair, nail type, and preferences.
2. Scan or upload images of products to retrieve detailed information on ingredients, usage, and safety.
3. Check individual ingredients for safety, allergen potential, and suitability for the consumer’s profile.
4. Provide AI-driven personalised routines for skincare, haircare, nail care, makeup, fragrances, and men’s grooming products.
5. Offer AR makeup try-on features for foundation, lipstick, eyeshadow, and other cosmetic products.
6. Compare two or more products based on multiple criteria (ingredient safety, brand reputation, price, eco-friendliness, effectiveness).
7. Filter products by type, sustainability, cruelty-free status, vegan options, or recyclable packaging.
8. Display match percentages to show how suitable a product is for a consumer’s skin/hair/nail type.
9. Enable consumers to share routines, reviews, and “look of the day” posts within the community section.
10. Gamification features: quizzes, challenges, streaks, and badges.
11. Dermatologist Mode: allow consumers to share scans/routines with verified dermatologists for professional guidance.
12. AI skin coach: daily check-ins and advice based on consumer data or sensor input.
13. Support integration with smart sensors or wearable devices for real-time skin/hair monitoring, allowing consumers to either sync their own devices or purchase compatible devices directly from the platform.
14. Allow consumers to purchase skincare, makeup, haircare, nail care, fragrances, and men’s grooming products directly through the app.

Non-Functional Requirements:

1. Fast response times for AI analysis, AR rendering, and product searches.
2. Ability to handle large numbers of consumers, product entries, and AI computations.
3. Ensure high availability with minimal downtime and implement robust error handling.

4. Protect sensitive consumer data, images, and personal information to comply with GDPR and other relevant regulations.
5. Design intuitive consumer interface suitable for a broad age range.
6. Use a modular design to allow easy updates to product databases, AI algorithms, and any other app feature.
7. Ensure compatability with different mobile operating systems (iOS and Android), and support for a wide range of device screen resolutions.
8. Consider features for visually impaired consumers or other accessibility needs.

Requirements Elicitation

8. Could observation of exisitng processes or behaviours (Ethnography) be used in this case study? If so, in what way?
   Yes, observing how people currently use skincare, makeup, haircare, nail care, fragrances, and men's grooming products could really help design the app. By watching how people shop, choose products, follow routines, or share tips with friends, we can see what works and what's frustrating. For example, we could notice how hard it is to pick the right foundation shade or how consistently they stick to their routines. These insights could then help tailor features in PerfectMatch, like personalised routines, AR try-ons, or daily reminders, making sure the app actually fits how people behave in real life.

9. Identify one or two significant stakeholder(s), which will be **interviewed** to get more information on the intended product.
   Justify your choice of stakeholder(s).
   Do up an interview plan.
   Prepare approximately 10 questions if interviewing 1 stakeholder or 6 to 8 questions if interviewing 2 stakeholders.
   STAKEHOLDERS
   1. Dermatologists can provide professional insight on how AI-generated routines and skin analysis tools should work to ensure safety and effectiveness. They can also advise on medical accuracy and help identify potential risks.
   2. General consumers can provide feedback on usability, features they would find helpful, common struggles when buying products, and community engagement expectations. Since they are the actual target audience, it's fundamental to ensures the app meets their real needs and preferences.

   INTERVIEW PLAN
   If I were to organise interviews, I would keep them short (10–20 minutes each) and run them either in person or online through Zoom/Teams. To encourage people to take part, I would offer a small incentive, such as entering them into a draw to win a voucher. I would also create an online form and share it on social media to reach more people and collect as much feedback as possible.

QUESTIONS

1. Dermatologists
    - What do you think are the most important things to consider when recommending a skincare/haircare routine?
    - Do you think AI skin analysis could be reliable? What could be the risks and limitations?
    - What should the "Dermatologist mode" inlcude to actually be useful?
    - Are there any ingredients that always need to be flagged for the consumers' safety?
    - How could the app add value to what dermatologists already do without trying to take their place?
    - How often do you think people should change or update their skincare routine?
2. General consumers/consumers
    - What's the hardest part for you when buying skincare/makeup?
    - Do you currently use any apps or websites to help you pick products or build routines?
    - Which features would you find the most helpful in PerfectMatch?
    - Would you feel comfortable uploadinf selfies or personal data to the app? Why or why not?
    - How much does sustainability matter when choosing products?
    - Would you actually buy products directly from the app if that option existed?

10. Identify a significant group of stakeholders, which will receive **questionnaires**.
    Justify your choice of stakeholders.
    The questionnaire should have approximately 10 questions.
    For the questionnaire, I think the most significant group of stakeholders would be general consumers. They are the main target audience of PerfectMatch, so their feedback is essential to understand what features they value, what problems they face when buying or using beauty products, and how likely they are to use an app like this.

    QUESTIONNAIRE

    1. What age group do you fall into?
        - Under 18
        - 18-24
        - 25-34
        - 35-44
        - 45+
    2. How often do you buy skincare/makeup/haircare/etc. products?
        - Weekly
        - Monthly
        - Every few months
        - Rareley
    3. Where do you usually shop for these products?
        - Online (brand websites, Amazon, etc.)

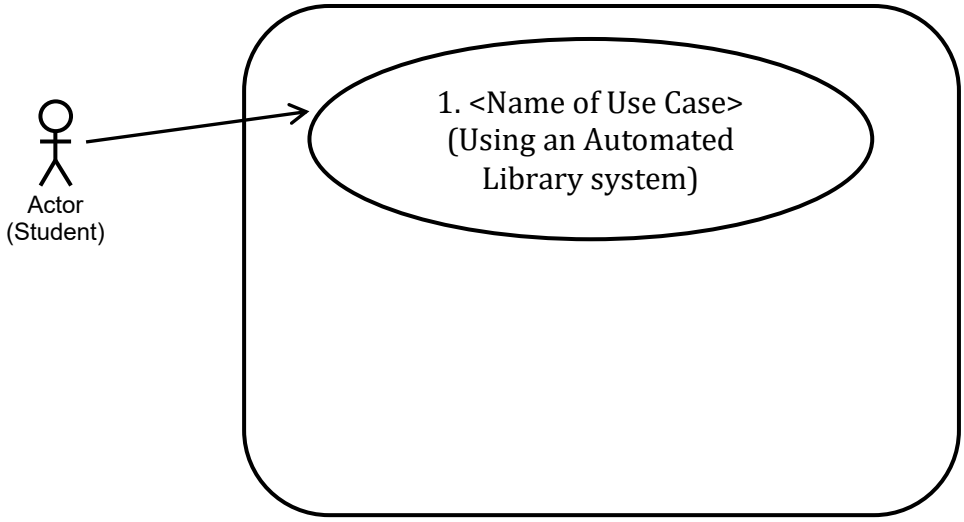
    - In-store (pharamacy, beauty store, supermarket, etc.)
    - Both equally
4. What do you usually find most difficult when buying new products?
    - Choosing the right product for my needs
    - Understanding ingredients safety
    - Sticking to a routine
    - Price/affordability
    - Other (please specify)
5. How important are eco-friendly, cruelty-free, or vegan options to you when buying products?
    - Very important
    - Somewhat important
    - Not important
6. Would you be interested in using an app that analyses your skin (through photos or sensors) and suggest personalised routines?
    - Yes
    - Maybe
    - No
7. How comfortable would you feel uploading selfies or routine data to an app like this?
    - Very comfortable
    - Somewhat comfortable
    - Not comfortable
8. Which features would you find most useful in the app? (Select all that apply)
    - Ingredient checker
    - AR makeup try-on
    - Product comparison
    - Personalised routine builder
    - Darmatologist mode
    - Community features (sharing routines, reviews, etc.)
9. Would gamification features (e.g. streaks, badges, or daily challenges) motivate you to use the app regularly?
    - Yes
    - Maybe
    - No
10. If this app allowed you to buy products directly, how likely would you be to shop through it?
    - Very likely
    - Somewhat likely
    - Not likely

Requirements Analysis

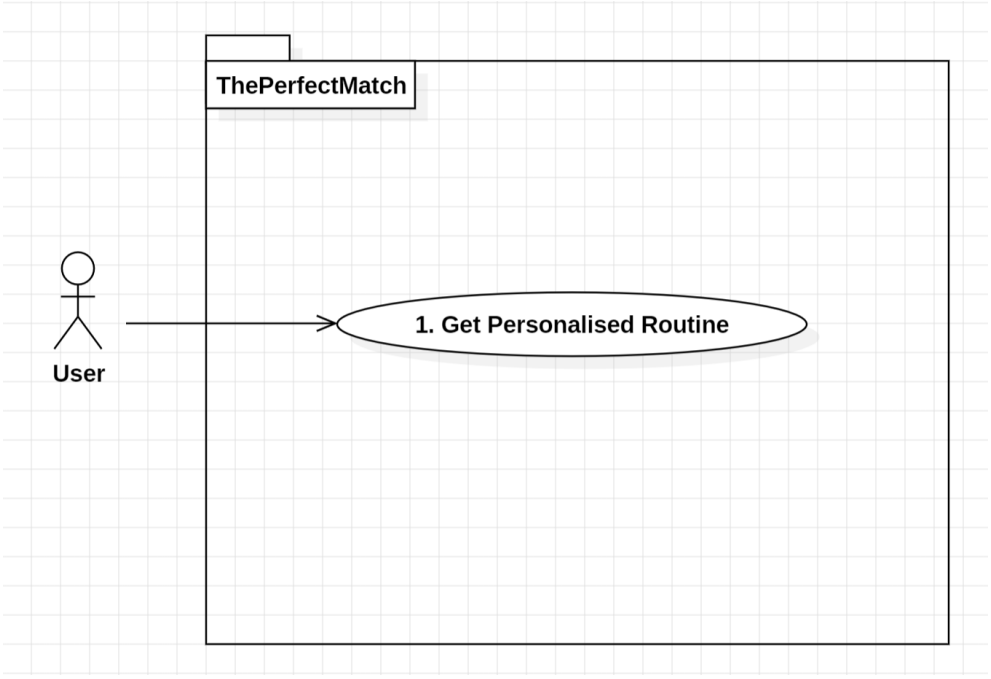
11. Use the use case template to analyse the proposed system.

Draw an initial *use-case diagram* with supporting scenario description for this app (possibly using *StarUML* for the diagram).

The first iteration of the use-case diagram can consist of a single overall use case with supporting main flow and 2 or 3 alternative flows.



System Name
(Automated Library System)



The use case description is developed from analysing the description of the use case. This is the long description of the use case, in the use case narrative.
For the first iteration this will be a description of the how the system operates.
Use Cases focus on functional requirements and specific system behaviour.

USE CASE	Use Case: 1	Get Personalised Routine
Brief Description of Goal in Context	The consumer wants to receive a personalised self-care routine, covering skincare, haircare, makeup or grooming. The system analyses the consumer's inputs such as selfies, quiz answers, or smart sensor readings and produces a tailored routine that fits their profile, preferences, and goals.	
Preconditions	- The consumer has installed ThePerfectMatch app and created an account. - The consumer has provided necessary personal data (e.g. skin type, allergies, goals, preferences)	
Post Conditions, Success End Condition	- A personalised routine is generated and displayed to the consumer. - Consumer can save, share, or view product suggeestions related to the routine.	
Long DESCRIPTION of Scenario	The consumer opens ThePerfectMatch app and selects the option to get a personalized routine. The system requests input such as a selfie, answers to a few lifestyle or skin-related questions, or optional sensor data. It analyses the information received using AI algorithms and a product database. Based on the consumer's profile and preferences, the system created a customised routine that includes recommended products and step-by-step guidance. The consumer views the routine, and may save or share it for later use.	

Main Flow		
Step	**Action**	**Alternate**
n.1.1	The consumer opens ThePerfectMatch app.	
n.1.2	The system displays the main dashboard.	
n.1.3	The consumer selects "Get Personalised Routine".	
n.1.4	The system presents available input options (selfie, quiz, sensor).	
n.1.5	The consumer chooses one input method.	
n.1.6	The system activates the selected input preference.	
n.1.7	The consumer provides required data (image, answers, or readings).	E.F. 1.7
n.1.8	The system validates the received data.	
n.1.9	The system analyses the data using AI algorithms.	
n.1.10	The system retrieves relevant products from the product database.	E.F. 1.10
n.1.11	The system matches products to consumer preferences and restrictions.	
n.1.12	The system generates a complete personalised routine.	
n.1.13	The system displays the routine with step-by-step guidance.	
n.1.14	The consumer reveiws the routine.	
n.1.15	The consumer saves the personalised routine for future use.	
n.1.16	End Use Case.	

EXCEPTIONS or ERROR Flow Description		
Step	**Branching Action**	**Alternate**

	< Exception number m of Use Case n>	
E.F. 1.7.1	Selfie upload fails.	
E.F. 1.7.2	Consumer retries.	
E.F. 1.7.3	Consumer chooses another option.	
E.F. 1.7.4	Go to Main Flow 1.6	
E.F. 1.10.1	No product found.	
E.F. 1.10.2	System asks to broaden filters.	
E.F. 1.10.3	Go to Main Flow 1.7	
ALTERNATIVE or VARIATION Flow Description		
Step	**Branching Action** No alternative flows as no options given in Use Case.	**Alternate**

Non Functional Requirements

Non-functional Requirements for each use Case can be added in the Table below.

Consider 2 or 3 non-functional requirements, related to the use case, from the list of options and add them to the table.
Non-functional requirements, management issues and decisions required to be made, can be identified in the following table.

From the table below, **choose <u>a limited number</u> of appropriate non-functional requirements** relevant to the Use Case.

Non Functional Requirements can be categorised as

- Product related
- Organisation related, process and approaches set by the company
- External, imposed by outside bodies

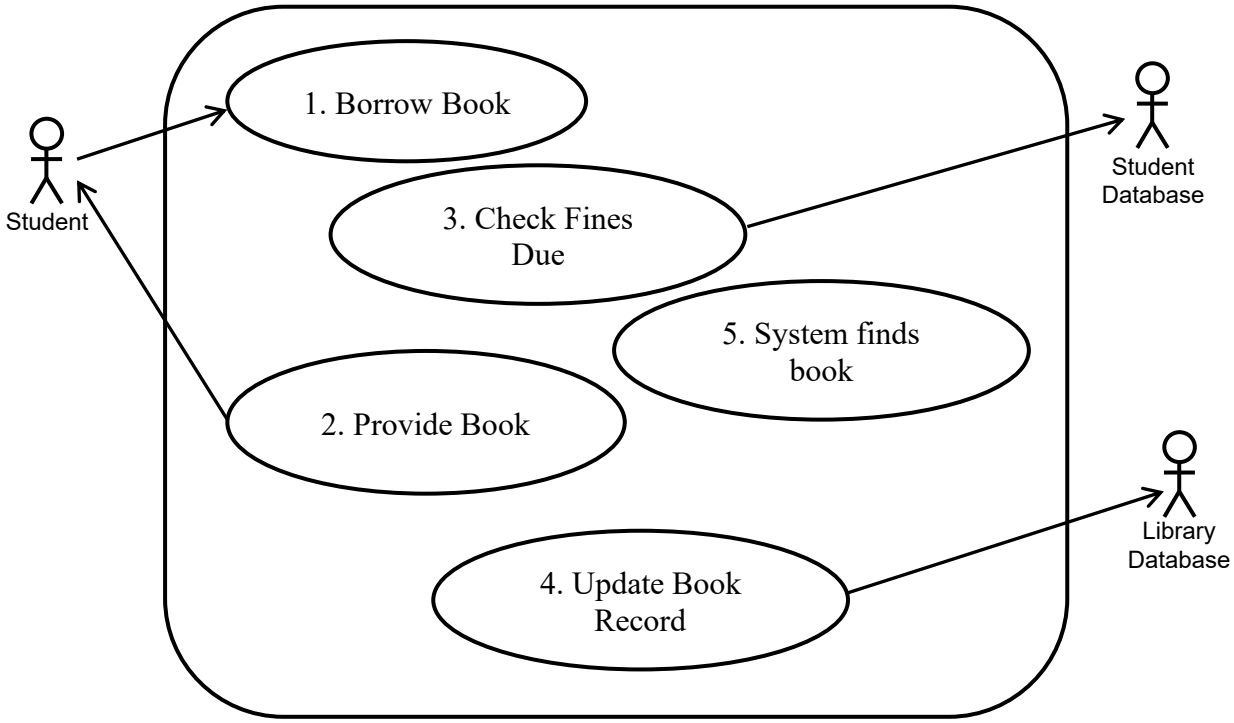
RELATED INFORMATION	Use Case: 1	Get Personalised Routine
Priority:	High – most important features of the app.	
Product: Performance	AI routine generation must complete within 5 seconds. [1].	
Product: Efficiency	The app should not be too large (under 150MB) [2]. It should not drain battery too quickly.	
Product: Reliability	The app should be available almost all the time (99.9% uptime) [3]. It can have a down time of 1 hour per month.	
External: Legislation	All consumer data (selfies, health data) must comply with GDPR [4].	
External: Ethical	The app must show a disclaimer, e.g. “Do not consider this app as a doctor.” It should avoid algorithmic bias.	
Channels to actors	Consumer interacts through mobile app UI. Sensors connect via Bluetooth.	
OPEN ISSUES	1. Source of verified product data. 2. Integration of smart wearables.	
Due Date	Dec 2025	

…any other management information…	

REFERENCES
[1] “Virtual Try-On Technology Guide”, 2025, Andrew Makarov, Mobidev, https://mobidev.biz/blog/augmented-reality-virtual-try-on-technology-for-ecommerce
[2] “Optimize your app’s size and stay within Google Play app size limits”, 2025, Google Help, https://support.google.com/googleplay/android-developer/answer/9859372?hl=en
[3] “The Rule of Nines: Why You Need 99.999% Availability in an Always-On World”, Penguin Solutions, https://www.penguinsolutions.com/en-us/resources/blog/rule-nines-availability-always-on-world#:~:text=What%20does%2099%25%20uptime%20mean,of%20unplanned%20downtime%20per%20year.
[4] “GDPR Compliance for Apps: A 2025 Guide”, 2025, Ana Mishova, GDPR Local, https://gdprlocal.com/gdpr-compliance-for-apps/

More Systems Analysis

12. Open a new report document for the **Systems Modelling.**
    Develop a second iteration in a separate word report consisting of 4 or 5 use cases.
    Each use case requires a use case narrative describing the scenario analysis.
    Each narrative requires to have at least 6 to 10 steps in the main flow.
    Each use case should have 2 or 3 exception or alternative flows



Automated Library System

Requirements Specification Matrix

13. From the requirements analysis identified with the use Case analysis identify key *functional* requirements.

    There should be 6 to 10 easily identifiable features or requirements that can be listed in this matrix.
    All the features (that will become use cases) identified previously need to be included, review section 7 for the additional features.

Req ID	Name of Req	Description	Priority	Consumer Contact
FR1	Get Personalised Routine	The app analyses the consumer's selfie, quiz answers, or sensor data to generate a custom routine with verified products suited to their skin, hair, or grooming needs.	High	Consumer interacts through mobile app; inputs selfie, quiz, or sensor data.
FR2	Get Expert Feedback	Allows verified dermatologists to review AI-generated routines and provide professional feedback or corrections for safety and accuracy.	High	Consumer shares routine with dermatologis t; receives notifications and feedback.
FR3	AR Try-On	Lets consumers virtually test makeup or grooming products (e.g. lipstick, foundation, hair dye) using the device's camera and AR overlay.	High	Consumer uses phone camera through the app's AR interface.
FR4	Process Consumer Data	System analyses selfies, quiz answers, and wearable data to detect skin or hair type, and stores results for personalisation.	High	Consumer uploads or provides real-time data via app and sensors.
FR5	Supply Product Data	Provides accurate product details such as ingredients, safety ratings, price, and sustainability data through a verified database.	High	Consumer queries database via search or scanner feature.

FR6	Purchase Products	Enables consumers to purchase verified products directly through the app using a secure payment gateway and get digital receipts.	High	Consumer selects “Buy Now” on product page; payment processed via secure gateway.
FR7	Provide Skin Coaching	Sends daily AI-generated skin advice (e.g. “Use lighter moisturiser today”) based on personal data and environmental factors.	Medium	Consumer receives push notifications or in-app coaching tips.
FR8	Check Ingredient Safety	Analyses product ingredients for allergens, toxins, or restricted substances, providing a safety report and safer alternatives.	High	Consumer scans barcode or uploads ingredient list for instant feedback.
FR9	Compare Products	Compares two or more products by price, safety, eco-rating, and compatibility, showing a ranked recommendation table.	Medium	Consumer selects multiple products within the app for side-by-side comparison.
FR10	Create Community Content	Influencers or regular users can post skincare routines, product reviews, and advice to the app’s community feed.	Medium	Consumer uploads text, images, or videos to community section for interaction.

System Modeling

From the systems analysis and the requirements table, identify additional features and actors.
Normally this would be done through additonal Use Case models (diagrams and narratives).
At this stage the aim is just to list what would be needed to complete the model in a list by reviewing the requirements table and the systems analysis models.

14. List all/most of the potential *actors* in this system.
    - Consumer: The main user of the app who checks ingredients, builds routines, compares products, and tries makeup virtually.
    - Dermatologist: A verified professional who reviews AI-generated routines and provides expert feedback through the "Dermatologist Mode".
    - Influencer: Users who create or share content like routines, reviews, or beauty tips in the community section.
    - System Administrator: Manages the app, updates databases, ensures data security, and handles user or system issues.
    - AI Engine: The internal system component that analyses selfies, quiz data, or sensor input to generate personalised routines.
    - Product Database: The verified database that supplies product details (ingredients, safety info, pricing, etc.) to both the AI and consumers.
    - Payment Gateway: A secure third-party service that processes purchases within the app.
    - Smart Sensors: External devices like smartwatches that send data about skin condition or environment.

15. List many more of the potential *use-cases* in this system.
    - Get Personalised Routine: User gets a full skincare, makeup, or grooming routine based on their profile, selfies, or sensor data.
    - Get Expert Feedback: The dermatologist reviews and adjusts AI-generated routines for accuracy and safety.
    - AR Try-On: User virtually tests products like foundation or lipstick using the device camera.
    - Process Consumer Data: System analyses selfies or quiz inputs to detect skin type, issues, and preferences.
    - Supply Product Data: Database provides verified info on product ingredients, safety, and eco-ratings.
    - Purchase Products: The user can buy recommended or selected products directly through the app using a secure payment gateway.
    - Provide Skin Coaching: The app gives daily personalised advice like "Use extra moisturiser today" or "Avoid exfoliating tonight."
    - Check Ingredient Safety: The user scans a product barcode or uploads an ingredient list, and the system checks for allergens, toxins, or irritants.
    - Compare Products: The user can pick two or more products and compare them side by side based on factors like price, eco-rating, ingredient safety, and brand reputation.

- Create Community Content: Influencers or regular users can post skincare routines, product reviews, or educational content to the app's community feed.
- Join Challenges or Quizzes: Gamified features where users complete skincare challenges or quizzes to earn badges.
- Follow Dermatologists or Influencers: User can subscribe to verified experts for personalised tips.
- Manage Profile and Preferences: Edit skin type, goals, allergies, or routine preferences.
- Report or Flag Inappropriate Content: Community safety feature for moderation.
- Receive Notifications: Daily reminders, expert feedback alerts, or community updates.
- Sync with Smart Devices: Connect wearables to collect environmental and skin data in real-time.
- View Community Feed: Users browse other people's posts, comments, or like them.
- Capture Consumer Data: Collects selfies, quiz responses, and sensor input from the user before AI analysis.
- Add to Cart: Allows the consumer to select and store products in a virtual cart before checkout.
- Process Payment: Handles secure transactions using verified payment gateways.
- Generate Receipt: Creates and stores a digital receipt after a successful purchase.

Validation and Verification of Requirements

16. Test Case Planning.

    Develop test cases for the main use cases, the abstracted **high priority** functional requirements, identified in Iteration 2 or 3 of the Use Case Analysis.

    Develop at least 1 test case for the **most important** non-functional requirements for the highest priority Use Cases.

    Use the test case template to create initial Use Acceptance Test plans that will permit consumers and developers to agree the system will have been developed as specified by the requirements

    Consider the test plan as a consumer guide or consumer manual for non-technical novice consumers of the system

Test Case Number: 1
Test Case Name: Verify Get Personalised Routine
Related Use Case Name: Get Personalised Routine Number: **1**
Purpose: **Confirm that the system correctly generates a personalised routine.**
Procedure Steps: (Guided by Main flow or other flows of Use case) 1. Tester opens the PerfectMatch app. 2. Tester logs in using a valid account. 3. Tester selects "Get Personalised Routine." 4. System prompts for input options (selfie, quiz, or sensor data). 5. Tester uploads a valid selfie and answers quiz questions. 6. System validates input and runs AI analysis. 7. System retrieves verified products from the database. 8. Personalised routine is generated. 9. Tester views the suggested products and saves the routine.
Expected Results: **Routine is displayed with relevant product suggestions and stored in the tester's account. All steps complete smoothly without errors.**

Test Case Number: 2

Test Case Name: Verify Get Expert Feedback

Related Use Case
Name: Get Expert Feedback
Number: **2**

Purpose:
Ensure that dermatologist feedback is correctly submitted, stored, and viewable by the consumer.

Procedure Steps: (Guided by Main flow or other flows of Use case)

1. Tester (acting as a dermatologist) logs into the professional dashboard.
2. System displays pending consumer routine for review.
3. Tester opens the routine and adds comments or edits.
4. Tester submits feedback.
5. System saves the feedback securely and sends notification to the consumer.

Expected Results:
Consumer receives notification about new expert feedback. Feedback is visible under the saved routine section.

Test Case Number: 3

Test Case Name: Verify AR Try-On

Related Use Case
Name: AR Try-On
Number: **3**

Purpose:
Verify that the AR system displays realistic product previews using the device camera.

Procedure Steps: (Guided by Main flow or other flows of Use case)

1. Tester selects “AR Try-On” from the dashboard.
2. System requests camera permission.
3. Tester grants permission.
4. Tester chooses a product (e.g. lipstick shade).
5. System overlays product on live camera feed.
6. Tester adjusts intensity and saves preview.

Expected Results:
AR effect appears in real-time, matches product colour accurately, and saves successfully.

Test Case Number: 4

Test Case Name: Verify Process Consumer Data

Related Use Case
Name: Process Consumer Data
Number: **4**

Purpose:
Confirm that the system accurately analyses consumer inputs and stores structured data in the user profile.

Procedure Steps: (Guided by Main flow or other flows of Use case)

1. Tester opens the PerfectMatch app and logs in.
2. Tester selects "Get Personalised Routine" and provides a selfie or quiz data.
3. The system validates the input and ensures completeness.
4. The AI module analyses skin/hair features using trained models.
5. The processed data (e.g., skin type, dryness, acne level) is structured and stored in the user's profile.
6. Tester checks that the stored data appears correctly in the user profile.

Expected Results:
The system correctly processes data and stores structured, accurate results within 5 seconds. Errors (e.g. missing data) trigger a retry prompt.

Test Case Number: 5

Test Case Name: Verify Supply Product Data

Related Use Case
Name: Supply Product Data
Number: **5**

Purpose:
Confirm that the system retrieves and displays verified product information from the product database accurately.

Procedure Steps: (Guided by Main flow or other flows of Use case)

1. Tester opens the PerfectMatch app.
2. Tester navigates to the product search or routine recommendations section.
3. Tester selects a product or enters a product name in the search bar.
4. The system sends a query to the Product Database.
5. The database retrieves and filters relevant product information.
6. The system displays verified product details (ingredients, price, eco-rating, brand, etc.).
7. Tester checks that all data fields are complete and match database entries.

Expected Results: **The system displays full, accurate product information within 3 seconds of the search request. Missing or outdated data triggers an automatic refresh or error notice.**

Non-Functional Test Case

Test Case Number: N/A
Test Case Name: Verify System Performance and Responsiveness
Related Requirement Name: Performance
Purpose: **Ensure the app meets the defined response time and reliability targets for key actions.**
Procedure Steps: (Guided by Main flow or other flows of Use case) 1. Tester measures AI analysis response time for generating a routine. 2. Tester checks AR Try-On startup delay. 3. Tester monitors app performance under normal load.
Expected Results: • **AI routine generated in under 5 seconds.** • **AR Try-On loads within 2 seconds.** • **App remains stable with no crashes during use.** • **Availability meets 99.9% uptime requirement.**

Completing the Feasibility Study

Review Previous Versions

17. Before you complete the final submission of a feasibility report, review and update the non-functional requirements, if necessary.

18. Review and, if necessary, Update the requirements Specification Matrix, section 13, and identify the high level (abstract) core system features.

Update Requirement Specification (RS) & set of Use-Case Diagrams (UCD) with narratives

19. Consider the Use case Model to ensure that core/key functionality has been addressed in the analysis and modelling process.

    Add comments here on what might need to be done to address any omissions or corrections.

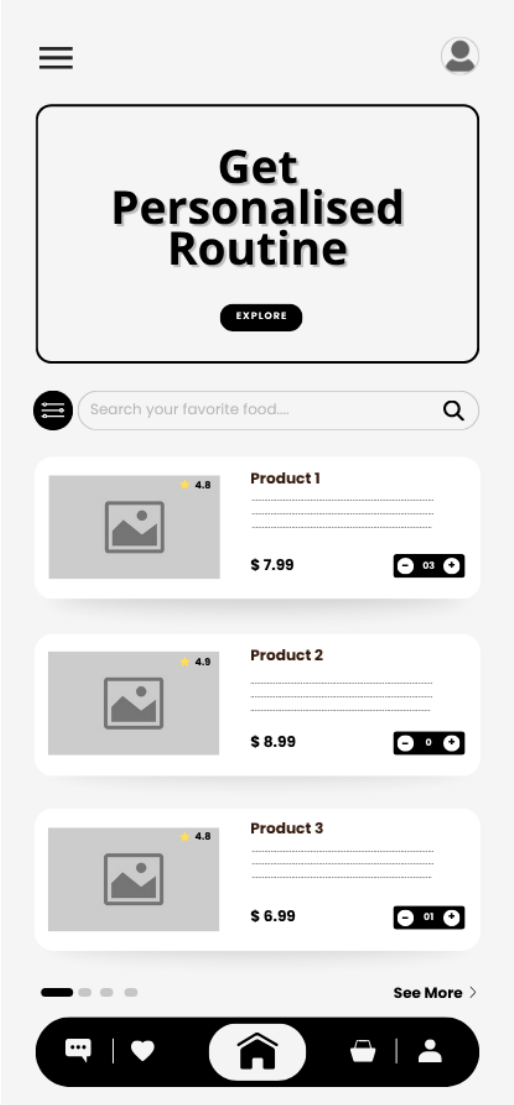
    a. Do any of your use-cases need to be broken down further i.e., is there is too much functionality in one use-case?
    1. Get Personalised Routine: Since this use case currently combines multiple actions (AI analysis, data collection, and product recommendation), it could be separated into sub-use-cases such as "Capture Consumer Data" and "Generate Routine".
    2. Purchase Products: This could be split into "Add to Cart", "Process Payment", and "Generate Receipt".
    b. Update the potential Use Case list in section 15 as necessary.
    c. Update the Requirement Specification table with additional requirements as necessary.

Prototype

20. Create an initial prototype of the proposed system.

    Such as: sketch the home page/starting page of the system. Then take a photo of it and insert the photo into the document.

    Alternatively use a drawing tool to design the prototype view of the home page of the app.



Functional & Non-Functional Test-Cases

21. Write 3 additional test-cases (using the test-case template) for each of three abstracted **high priority** *functional* requirements (one test-case per requirement/use case).

Test Case Number: 6
Test Case Name: Verify Purchase Products
Related Use Case Name: Purchase Products Number: **6**
Purpose: **Test the secure in-app purchasing and payment flow.**
Procedure Steps: (Guided by Main flow or other flows of Use case) 1. Tester opens a product page within the app. 2. Tester clicks "Buy Now." 3. System displays verified retailer info and checkout page. 4. Tester enters valid payment details. 5. System processes payment and confirms order. 6. Receipt is saved to the tester's account.
Expected Results: **Payment completes successfully, confirmation screen appears, and receipt is stored in purchase history.**

Test Case Number: 7
Test Case Name: Verify Provide Skin Coaching
Related Use Case Name: Provide Skin Coaching Number: **7**
Purpose: **Ensure that the app successfully analyses recent user data and generates accurate daily skin coaching advice.**
Procedure Steps: (Guided by Main flow or other flows of Use case) 1. Tester logs into the app using a valid account. 2. Tester enables the "Skin Coaching" feature in settings. 3. System retrieves stored consumer profile and recent skincare usage data. 4. System analyses data for trends or inconsistencies. 5. System generates personalised advice based on user habits and environmental conditions. 6. The message appears on the dashboard and sends a push notification.

Expected Results: **The consumer receives personalised advice (e.g. "Use extra moisturiser today") within 5 seconds. The notification matches the user's data, and the advice is stored in the history section for review.**

Test Case Number: 8
Test Case Name: Verify Check Ingredient Safety
Related Use Case Name: Check Ingredient Safety Number: **8**
Purpose: **Validate that the system accurately detects unsafe or allergenic ingredients.**
Procedure Steps: (Guided by Main flow or other flows of Use case) 1. Tester scans a product barcode. 2. System reads and extracts the ingredient list. 3. System checks ingredients against the verified safety database. 4. System highlights allergens or unsafe substances. 5. Tester views the safety report and suggested alternatives.
Expected Results: **Ingredient report displays correctly, with warnings or safe/unsafe labels visible to the tester.**

22. Write 2 additional test-cases (using the test-case template) for each of the two **most important** *non-functional* requirements (one test-case per requirement).

Test Case Number: N/A
Test Case Name: Verify Fast Response Times for AI and AR Features
Related Requirement Name: Performance
Purpose: **Ensure the system meets the required response times for AI analysis, AR rendering, and product searches so consumers experience smooth and efficient app performance.**

Procedure Steps: (Guided by Main flow or other flows of Use case)

1. Tester logs into the PerfectMatch app.
2. Tester starts the Get Personalised Routine process by uploading a selfie.
3. Tester records the time taken for the AI analysis and routine generation to complete.
4. Tester opens AR Try-On and measures the time from camera activation to AR overlay display.
5. Tester searches for a product using the search bar and records how long results take to appear.
6. Tester repeats each test three times under normal network conditions to confirm consistency.

Expected Results:

- **AI routine generated in under 5 seconds.**
- **AR Try-On loads within 2 seconds.**
- **App remains stable with no crashes during use.**
- **Availability meets 99.9% uptime requirement.**

Test Case Number: N/A

Test Case Name: Verify Consumer Data Protection and GDPR Compliance

Related Requirement
Name: Security & Privacy

Purpose:

Ensure the app protects all sensitive consumer data (e.g. selfies, quiz responses, and skin analysis results) in compliance with GDPR, including consent handling, encryption, and secure data deletion.

Procedure Steps: (Guided by Main flow or other flows of Use case)

1. Tester opens the PerfectMatch app and registers a new account.
2. Tester uploads a selfie and completes the skincare quiz.
3. System requests explicit consent for data storage and processing (as per GDPR).
4. Tester verifies that the privacy notice and consent options are displayed before submission.
5. Tester inspects network activity to confirm that transmitted data is encrypted (HTTPS).
6. Tester requests account deletion and verifies that all personal data (including images and analysis) is fully removed from the system.

Expected Results:

- **System displays privacy and consent notice before any data collection.**
- **Data transmission and storage use encryption (HTTPS / secure storage).**

- **Consumer can successfully withdraw consent and delete all stored personal data.**
- **System complies fully with GDPR privacy standards.**